

IISER Mohali -PHY102-Electromagnetism

Assignment 2b

Due date: 1st March 2025

Q1. Consider an electric dipole is formed with charges of " $20 \times 10^{-19} C$ " and " $50 \times 10^{-19} C$ ". Distance between two charges is "10 micron" and placed along "x-axis". Draw the potential along x-axis and electric field intensity curve along y-axis when distance r is varied from -5cm to 5cm along the y axis". while varying r, it crosses origin 0,0 point of the x-y co-ordinate. explain your answer with correct arguments.

Q2. Calculate divergence and curl of the given Electric field $E = 6xyi + (3x^2 - 3y^2)j$.

Q3. We know that $-\vec{\nabla}V = \vec{E}$ and $\vec{E}$ can be write for spherical homogeneous domain as $\vec{E} = \frac{qr}{4\pi\epsilon r^2} \hat{r}$, where $\hat{r}$ is an unit vector. Can you drive the form of "V" using above equations only.

Q4. Considering the situation of question number 3 (Q3), Prove $\vec{\nabla} \times \vec{E} = 0$

Q5. Consider a case of discharge/charging of the capacitor in RC circuit. Show the relationship between rate of change of charge density (ρ) and the current density (J)